P.4 Linear Equations in Two Variables

Name _____ Period ____ Date _____

Pre-Calculus P.4 (30-minute plan) · Student sheet

I can…

- find the slope of a line.
- write the equation of a line from a point and a slope or two points.
- write equations of parallel and perpendicular lines.

Vocabulary · fill in the missing word

Slope — the ratio of the _____ in y to the change in x

Slope-intercept form — y = _____

Point-slope form — $y - y_1$ = _____

General form — Ax + By + C = _____

Part 1 · Notes and examples (fill in each note, then work the example)

Slope: m = _____. Horizontal lines have slope 0; vertical lines have undefined slope.

Example 1

Find the slope through each pair.

a. (−2, 0), (3, 1)

b. (−1, 2), (2, 2)

c. (0, 4), (1, −1)

Point-slope form: $y - y_1$ = _____

Example 2

Write the equation of the line through (1, −2) with slope 3, in slope-intercept form.

Parallel lines have equal slopes; **perpendicular lines** have slopes that are _____ ($m_1 m_2 = -1$).

Example 3

Write the lines through (2, −1) that are (a) parallel and (b) perpendicular to 2x − 3y = 5.

Part 2 · Partner practice

Partner practice

Write the line through (8, 3) perpendicular to y = 4x − 1.